

A Lightweight and Robust Two-Stage Acoustic Pipeline for Embedded Artillery Launch Detection

Ing. Martin Maxa

Czech Technical University in Prague,
Faculty of Electrical Engineering
Prague, Czech Republic
maxamart@fel.cvut.cz

doc. Ing. Jakub Svatoš, Ph.D.

Czech Technical University in Prague,
Faculty of Electrical Engineering
Prague, Czech Republic
svatoja1@fel.cvut.cz

Bc. Jakub Zelinka

Czech Technical University in Prague,
Faculty of Electrical Engineering
Prague, Czech Republic
zelinjak@fel.cvut.cz

Abstract—Fielded acoustic artillery sensing must tolerate background noise and still run on low-power embedded devices. We study the post-trigger classification stage of a lightweight two-stage acoustic pipeline for artillery launch detection. The main corpus contains 706 clean labeled events: 62 artillery launches and 644 non-launch events. We train for robustness using an expanded non-launch corpus, class-imbalance mitigation, train-time augmentation, and we evaluate controlled MFCC domain perturbations reported as nominal SNR levels. Deployment on an ESP32-S3 with TensorFlow Lite Micro preserves strong classification performance under the tested conditions, reaching MCC values of 0.9912 at 30 dB and 0.8009 at 5 dB. The results show that a lightweight embedded post-trigger classifier is feasible within a broader two-stage detection pipeline, although the specific robustness effect of quantization is supported empirically rather than isolated as a causal mechanism.

Index Terms—acoustic detection, artillery launch, MFCC, CNN, TensorFlow Lite Micro, embedded AI, robustness, feature-space perturbation

I. INTRODUCTION

Passive acoustic sensing can support early warning in contested environments where active systems are expensive, power-demanding, or easier to locate. For artillery launch detection, a practical acoustic system must reconcile two constraints: (i) robustness to background noise and event variability and (ii) feasible execution on low-power embedded hardware.

We use a two-stage system view: a streaming trigger stage first identifies candidate impulsive events, and a post-trigger classifier then processes short acoustic segments. We treat the trigger, data acquisition, and circular-buffer logic as previously validated components and focus experimentally on the post-trigger artillery launch classifier. Within that scope, we test whether a lightweight classifier preserves useful discrimination under controlled perturbations, reported through nominal SNR levels, while still fitting deployment on ESP32-S3.

Our training strategy combines an expanded non-launch corpus, class-imbalance mitigation, and train-time augmentation, and we evaluate whether this design maintains near-clean behavior under increasing perturbation. We also assess whether the same lightweight classifier retains useful embedded behavior after quantization and deployment through TensorFlow Lite Micro on ESP32-S3, using PC-side int8 inference as the immediate reference point.

Specifically, we frame artillery-launch sensing as a two-stage problem and place the experimental focus on the post-trigger classifier. We then evaluate that classifier on an expanded 706-event corpus under controlled perturbation with nominal SNR reporting, and we report ESP32-S3 deployment results together with a PC-side int8 verification step.

II. STATE OF THE ART

A. Classical Acoustic Detection of Impulsive Events

Weapon-related acoustic surveillance has traditionally used thresholding, energy-envelope analysis, time-of-arrival logic, and multi-sensor consistency checks to detect impulsive transients before semantic classification. Operational pipelines usually split artillery and gunshot processing into two tasks: (i) event triggering in continuous audio streams and (ii) post-trigger interpretation of short signal segments. Battlefield-oriented studies showed that reliable operation depends on transient triggering that tolerates propagation distortion, changing stand-off distances, and heterogeneous sensor layouts [1]–[4]. Trigger quality therefore sets an upper bound on downstream classifier performance.

B. Acoustic Signature of Artillery and Gunfire

Artillery and gunfire recordings do not contain a single invariant waveform. Source type, firing geometry, propagation, and local reflections shape each event. Depending on scenario and range, a recording may include different mixtures of muzzle blast, projectile-related shock components, and delayed arrivals. Fig. 1 shows an example of a longer recorded artillery launch waveform before the 60 ms processing window is extracted. Earlier work captured these effects indirectly through time-frequency descriptors, while long-range studies modeled wave arrivals explicitly and documented sensitivity to atmosphere, terrain, and sensor geometry [1], [2], [4]. These factors create domain shift: models tuned on narrow or clean recording conditions often degrade when transferred to field recordings with stronger multipath, clutter, and nonstationary background noise.

C. From Detection to Post-Detection Classification

After trigger detection, systems usually classify short post-trigger windows. Classical approaches combine engi-



Fig. 1. Example waveform of a longer recorded artillery launch signal.

neered representations, such as wavelet, cepstral, or temporal-envelope features, with shallow classifiers. Later methods use CNN-style models over spectro-temporal inputs [3], [5]–[8]. Cepstral representations remain attractive in this setting because they compactly encode the spectral envelope of short acoustic events [9], while broader environmental-sound evidence also shows that complementary time-frequency descriptors can matter when transient temporal structure is informative [10]. Comparative evidence from related gunshot-classification experiments further suggests that performance depends on the interaction between feature family and classifier rather than on model choice alone; in particular, MFCC-based representations remained competitive against related cepstral alternatives in comparisons involving both neural and non-neural classifiers [11]. This favors a two-stage design with a robust trigger and a compact post-trigger classifier. In a small-data embedded setting, a compact representation can be preferable to richer inputs when robustness and deployment cost must be balanced together.

D. Robustness and Domain-Shift Handling

Recent studies report strong performance under uncontrolled recording conditions, but many results remain tied to dataset-specific priors, acquisition pipelines, and label granularity [5], [6], [11]. In practice, robustness failures often appear as performance collapse under perturbation, recording-device mismatch, background-scene changes, or the presence of overlapping sound sources [12]. This moves the design problem beyond clean-set accuracy. Augmentation strategy, perturbation tolerance, and operating-point calibration become central requirements [8]. For embedded artillery-launch sensing, one relevant question is whether class separability survives controlled perturbations applied to the feature representation even when the underlying recordings already come from real field conditions.

E. Embedded Deployment and Runtime Constraints

Edge deployment requires detection quality to be evaluated together with memory footprint, latency, and energy use. TinyML comparisons show that front-end feature extraction can dominate runtime, so the highest-accuracy model is not always the most deployable one [13]. Raspberry Pi and TPU-class studies further show that a moderate accuracy loss may be acceptable when real-time constraints and resource budgets are met [14]. Together with TensorFlow Lite deployment practice [15] and integer-only quantization methods for efficient on-device inference [16], these findings motivate validation

that separates desktop float32 behavior, quantized desktop verification, and embedded execution.

We therefore address a narrower question than full continuous surveillance: whether a lightweight post-trigger classifier for artillery launch sensing remains robust under controlled perturbation and still behaves well after quantized deployment on an embedded target.

III. METHOD

A. Processing Pipeline

We study the post-trigger classification stage of a two-stage acoustic pipeline. In the broader architecture, a streaming trigger identifies candidate impulsive events and forwards short buffered segments to a classifier. The trigger, acquisition chain, and online buffering logic are outside the scope of this study and are treated as previously validated components. We evaluate only the post-trigger artillery launch classifier.

Each candidate event is represented by a 60 ms waveform segment centered around the detected impulse peak. The segment is asymmetric: 40 % precedes the peak to capture onset structure, and 60 % follows it to retain decay and muzzle-blast content. Fig. 2 shows an example of the resulting waveform segment. MFCC features are then computed from these segmented signals for supervised CNN training and inference.

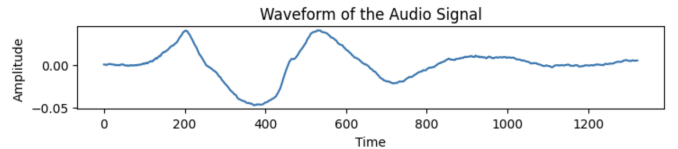


Fig. 2. Example 60 ms peak-centered waveform segment used for MFCC extraction.

B. Field Data Acquisition

The source corpus comes from real artillery measurements and includes both launch and impact acoustic events. We collected the recordings with the field station shown in Fig. 3. The acquisition chain used PreSonus PRM1 microphones connected through a Roland Rubix 44 audio interface, and signal processing uses 16-bit audio at 22.05 kHz. We model only binary artillery launch classification (artillery launch vs. non-launch) and leave impact-event modeling for future work. The positive class corresponds exclusively to artillery launch events, whereas the non-launch class contains other impulsive acoustic negatives.

The main clean corpus contains 706 labeled events: 62 artillery launches and 644 non-launch events. We expanded the non-launch class beyond the initial development corpus to better reflect operational negatives and to strengthen robustness-oriented training. An earlier, substantially smaller configuration with approximately 152 labeled events served only as a development-stage precursor and is not the main evaluation corpus reported here.



Fig. 3. Field recording station used for artillery dataset acquisition.

C. CNN Topology

The network input is an MFCC tensor of size $(58, 12, 1)$. The feature extractor has two Conv2D+MaxPool blocks (32 and 64 filters, kernel 3×3 , ReLU), followed by Flatten, Dense(64), Dropout(0.5), and a sigmoid output. Fig. 4 shows the full architecture. The final model is trained for 50 epochs with the Adam optimizer and binary cross-entropy loss. This layout balances representation capacity and deployment cost, following standard convolutional design practice [17].

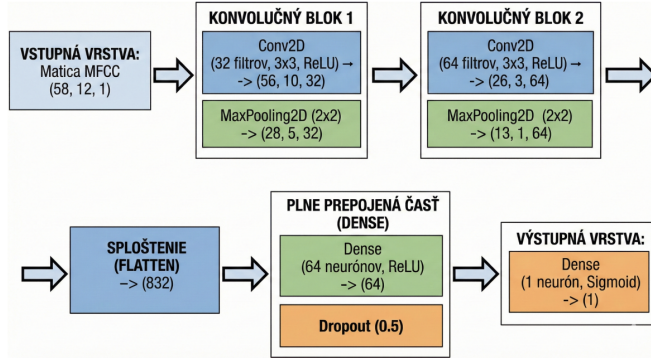


Fig. 4. CNN topology used for post-trigger artillery launch classification of MFCC segments.

D. Robustness-Oriented Training

MFCC extraction is performed on 22.05 kHz signals using short-time framing with high temporal resolution. Each 60 ms segment is divided into 2.5 ms frames (approximately 55 samples) with a 1.0 ms hop (approximately 22 samples), corresponding to 1.5 ms overlap (60 %). A Hamming window is applied to each frame to reduce spectral leakage, and the spectrum is evaluated with $NFFT = 512$. The mel front end uses 40 mel filters, from which the first 12 MFCC coefficients are retained. This yields the $(58, 12, 1)$ input tensor used by the classifier.

The main training strategy combines three elements: (i) an expanded non-launch corpus, (ii) class-imbalance mitigation, and (iii) train-time augmentation of the training split only. This follows the broader use of augmentation as a robustness mechanism in audio classification under limited-data conditions [8]. We apply perturbations directly to MFCC features to generate degraded variants, and we report nominal MFCC-space SNR levels of 30, 20, 10, and 5 dB. In the rest of the paper, we refer to these levels simply as nominal SNR. These values characterize controlled feature perturbation and should not be interpreted as waveform-domain acoustic SNR.

E. Imbalance and Overfitting Mitigation

Because the corpus is strongly imbalanced, with artillery-launch events representing only about 8.8 % of the 706 labeled samples, the training branch uses class-weighted optimization to reduce majority-class bias, following standard imbalanced-learning practice [18]. In practice, the loss assigns a stronger penalty to errors on the minority artillery-launch class so that high nominal accuracy cannot be achieved simply by collapsing predictions toward the majority non-launch class. We treat class weighting as part of the final robustness-oriented strategy, not as an isolated ablation result.

F. Embedded Inference Protocol

After PC training, we export the selected model to TensorFlow Lite Micro. To separate hardware effects from quantization effects, we evaluate three inference regimes: desktop float32 inference, desktop int8 inference with the quantized model, and ESP32-S3 deployment of the same quantized classifier. The embedded implementation uses int8 inference on ESP32-S3 with a statically allocated 80 KB `tensor_arena` for intermediate tensors and a selective `MicroMutableOpResolver` configuration that registers only the operators required by the network. This deployment choice follows standard integer-only inference practice for reducing memory footprint and enabling efficient execution on resource-constrained processors [16]. Embedded validation uses a host-device protocol: the host computes MFCC tensors, applies the same controlled perturbation settings used in desktop evaluation, transmits the tensors to the embedded target, and collects predicted labels and scores. This host-device transfer is a validation protocol rather than the intended operational pipeline. We use ESP32-S3 as a representative low-power Edge AI platform for classifier deployment, not as a proof of universal generalization across embedded hardware.

IV. EXPERIMENTAL SETUP

The main dataset contains 706 clean labeled events: 62 artillery launch and 644 non-launch events. We use a stratified event-level 80/20 train-test split, and any train-time augmentation is applied only after the split to prevent leakage between original events and their augmented variants. Robustness is evaluated under controlled perturbation at nominal SNR levels of 30, 20, 10, and 5 dB. We report accuracy, precision, recall,

F1-score, and Matthews correlation coefficient (MCC), and compare:

- 1) desktop float32 post-trigger classification under controlled MFCC-space perturbation,
- 2) PC-side int8 verification of the quantized model, and
- 3) ESP32-S3 deployment of the same quantized post-trigger classifier.

V. RESULTS

A. Post-Trigger Classification Under Nominal MFCC-Space SNR

Table I summarizes the desktop float32 reference. The model retains high recall across all tested SNR levels, but precision is low and degrades further under heavier corruption. As a result, MCC remains moderate and declines from 0.5377 at 30 dB to 0.4581 at 5 dB, indicating that the float32 reference is vulnerable mainly through false-positive inflation rather than through missed artillery launches.

TABLE I
DESKTOP FLOAT32 RESULTS UNDER CONTROLLED PERTURBATION

SNR (dB)	Accuracy (%)	Precision	Recall	F1-score	MCC
30	84.41	0.35	1.00	0.52	0.5377
20	82.86	0.33	1.00	0.49	0.5367
10	81.12	0.31	0.99	0.47	0.4893
5	80.44	0.29	0.93	0.45	0.4581

B. PC-Side int8 Verification

To separate hardware effects from quantization effects, we next evaluate the quantized model on PC before moving to the embedded target. The PC-side int8 results in Table II are consistently stronger than the float32 reference, with MCC values of 0.9211 at 30 dB and 0.6511 at 5 dB. This intermediate step suggests that quantization contributes to the robustness improvement observed later on the embedded target, although we do not isolate the exact mechanism from all preprocessing and deployment-related factors.

C. ESP32-S3 Deployment Results

Table II also reports the embedded ESP32-S3 results for the same quantized classifier. The deployment results preserve strong classification behavior across all tested SNR levels and do not show harmful drift relative to the PC-side int8 reference. In this setup, the embedded target attains MCC values from 0.9912 at 30 dB to 0.8009 at 5 dB. The average per-segment inference latency is approximately 32 ms, and the deployed implementation uses an 80 KB `tensor_arena`. These results indicate that the quantized post-trigger classifier remains deployable on the target platform under the tested conditions.

TABLE II
QUANTIZED INFERENCE COMPARISON: PC-SIDE INT8 VERIFICATION VS. ESP32-S3 DEPLOYMENT

Mode	SNR (dB)	Accuracy (%)	Precision	Recall	F1-score	MCC
PC int8	30	98.66	0.86	1.00	0.93	0.9211
PC int8	20	96.77	0.72	1.00	0.84	0.8340
PC int8	10	92.74	0.54	0.97	0.69	0.6890
PC int8	5	92.41	0.53	0.90	0.66	0.6511
ESP32-S3	30	99.86	1.00	0.98	0.99	0.9912
ESP32-S3	20	99.73	0.98	0.98	0.98	0.9824
ESP32-S3	10	97.78	0.86	0.87	0.87	0.8554
ESP32-S3	5	97.18	0.95	0.70	0.81	0.8009

VI. DISCUSSION AND LIMITATIONS

The results support a two-stage interpretation in which data acquisition and trigger generation act as upstream components, while the post-trigger classifier carries the robustness-critical semantic decision. The float32 desktop reference is not a sufficient proxy for deployment behavior under controlled perturbation. Quantized inference, first verified on PC and then deployed on ESP32-S3, stays substantially more stable under the tested nominal SNR levels.

The results also have clear limitations. First, we do not evaluate continuous-stream trigger performance or false-trigger rates, so our system-level claim is limited to the post-trigger classification stage. Second, the main evaluation uses a single stratified hold-out split; although we perform the split before augmentation to prevent leakage, broader uncertainty estimation is left for future work. Third, the robustness study is based on controlled perturbation in MFCC space, and the reported nominal SNR should not be interpreted as waveform-domain acoustic SNR. Fourth, the apparent robustness benefit associated with quantization is an empirical observation supported by the PC-side int8 bridge experiment, not an isolated mechanistic proof. Finally, the corpus remains relatively small in terms of positive artillery-launch events, and the deployment study covers ESP32-S3 rather than all embedded targets.

VII. CONCLUSION

We presented a lightweight two-stage acoustic pipeline for embedded artillery launch detection and evaluated its post-trigger classification stage under controlled perturbation. On the expanded 706-event corpus, the desktop float32 reference remains vulnerable to false positives, whereas quantized inference is markedly more robust. The PC-side int8 verification and ESP32-S3 deployment results show that the post-trigger classifier can run on the target platform without harmful deployment drift. Within the stated scope boundaries, these results support embedded post-trigger classification as a viable component of artillery-launch acoustic sensing.

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